

insertChar

Write the C function that takes in a string `str1` as an argument, copies the contents of character string `str1` into character string `str2`. In addition, the function also has a character parameter `ch`. For every three characters copied from `str1` to `str2`, the character `ch` is inserted into `str2`. The function returns the resultant string to the calling function via call by reference. For example, if the string `str1` is "abcdefg", and the inserted character `ch` is '#', then the resultant string `str2` = "abc#def#g" will be returned to the calling function. The function prototype is given as follows:

```
void insertChar(char *str1, char *str2, char ch);
```

A sample program template is given below to test the function:

```
#include <stdio.h>
#include <string.h>
void insertChar(char *str1, char *str2, char ch);
int main()
{
    char a[80],b[80];
    char ch, *p;

    printf("Enter a string: \n");
    fgets(a, 80, stdin);
    if (p=strchr(a, '\n')) *p = '\0';
    printf("Enter a character to be inserted: \n");
    ch = getchar();
    insertChar(a,b,ch);
    printf("insertChar(): ");
    puts(b);
    return 0;
}
void insertChar(char *str1, char *str2, char ch)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

- (1) Test Case 1:
Enter a string:
abc de
Enter a character to be inserted:

insertChar(): abc# de#
- (2) Test Case 2:
Enter a string:
abc
Enter a character to be inserted:

insertChar(): abc#
- (3) Test Case 3:
Enter a string:
I am a boy.
Enter a character to be inserted:
\$
insertChar(): I a\$m a\$ bo\$y.
- (4) Test Case 4:

```
Enter a string:  
hi  
Enter a character to be inserted:  
$  
insertChar(): hi
```